GeoModels Tutorial: analysis of spatio-temporal data with spatial locations changing over time using Gaussian random fields

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Introduction

In this tutorial we show how to analyze geo-referenced spatio-temporal data using Gaussian random fields (RFs) when the spatial coordinates change over time with the R package `GeoModels` [Bevilacqua et al., 2026].

We first load the R libraries needed for the analysis and set the name of the model in the `GeoModels` package:

```
rm(list=ls())
library(GeoModels)
library(fields)
model="Gaussian"  # model name in the GeoModels package
set.seed(881)
```

Simulation of a space-time Gaussian random field with spatial coordinates changing over time

Let us consider a space-time Gaussian RF $Z = \{Z(\boldsymbol{s}, t), \boldsymbol{s} \in \mathcal{S}, t \in \mathcal{B}\}$, where $\boldsymbol{s}$ represents a location in the domain $\mathcal{S}$ and t represents a temporal instant in the domain $\mathcal{B}$. We assume that Z is stationary with zero mean, unit variance and correlation function given by $\rho(\boldsymbol{h}, u) = \text{cor}\{Z(\boldsymbol{s}+\boldsymbol{h}, t+u), Z(\boldsymbol{s}, t)\}$.

Then we consider the RF $Y = \{Y(\boldsymbol{s}, t), \boldsymbol{s} \in \mathcal{S}, t \in \mathcal{B}\}$ defined by the location and scale transformation:

$$Y(\boldsymbol{s}, t) = \mu(\boldsymbol{s}, t) + \sigma Z(\boldsymbol{s}, t). \tag{1}$$

where $\mu(\boldsymbol{s}, t) = X(\boldsymbol{s}, t)^\top \boldsymbol{\beta}$ and $X(\boldsymbol{s}, t)$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_k)^\top$ is a k-dimensional vector of unknown parameters (in this tutorial we fix $k = 2$). Then $\text{E}\{Y(\boldsymbol{s}, t)\} = X(\boldsymbol{s}, t)^\top \boldsymbol{\beta}$, $\text{Var}\{Y(\boldsymbol{s}, t)\} = \sigma^2$ and $\text{Cov}\{Y(\boldsymbol{s}+\boldsymbol{h}, t+u), Y(\boldsymbol{s}, t)\} = \sigma^2 \rho(\boldsymbol{h}, u)$.

Suppose we want to simulate a realization of Y at $t_1 = 0, t_2 = 1, \ldots, t_T = 15$, $T = 16$ temporal instants and spatial locations (changing over time) $\boldsymbol{s}_{i(l)}$, with $i(l) = 1, \ldots, N_l$ and $l = 1, \ldots, T$, uniformly distributed in the unit square.

We first set the temporal instants and then the (changing over time) spatial coordinates with associated covariates.

```
coordt=seq(0,15,1)  # temporal instants
coordx_dyn=list()  # dynamical spatial coordinates
X=list()           # dynamical spatial covariates
minN=150; maxN=250
for(k in 1:length(coordt))
{
 NN=sample(minN:maxN,size=1)
 x=runif(NN,0,1)
 y=runif(NN,0,1)
 coordx_dyn[[k]]=cbind(x,y)
 X[[k]]=cbind(rep(1,NN),runif(NN))
}
```

Note that both the dynamical spatial coordinates and the covariates are saved as lists. The numbers of location sites $N_1, \ldots, N_{16}$ for each temporal instant are:

```
unlist(lapply(coordx_dyn,nrow))
[1] 191 243 214 216 208 155 184 210 202 179 205 225 222 244 209 166
```

and the total number of space-time locations is $N = \sum_{l=1}^{16} N_l = 3273$ in this example. The spatial coordinates for the first two temporal instants are depicted in Figure 1.

```
plot(coordx_dyn[[1]],pch=20)
plot(coordx_dyn[[2]],pch=20)
```

We then specify the mean, variance and nugget parameters:

```
mean=0.2; mean1=-0.3
sill=1; nugget=0
```

where `mean`, `mean1` and `sill` are respectively β_1, β_2 and σ^2. For the correlation function we assume a simple spatially isotropic and symmetric in time double exponential model:

$$\rho((\boldsymbol{h}, u); \alpha_s, \alpha_t) = \exp\left(-\frac{\|\boldsymbol{h}\|}{\alpha_s} - \frac{|u|}{\alpha_t}\right). \tag{2}$$

Then we set the name of the correlation model and the associated parameters and save all the parameters as a list.

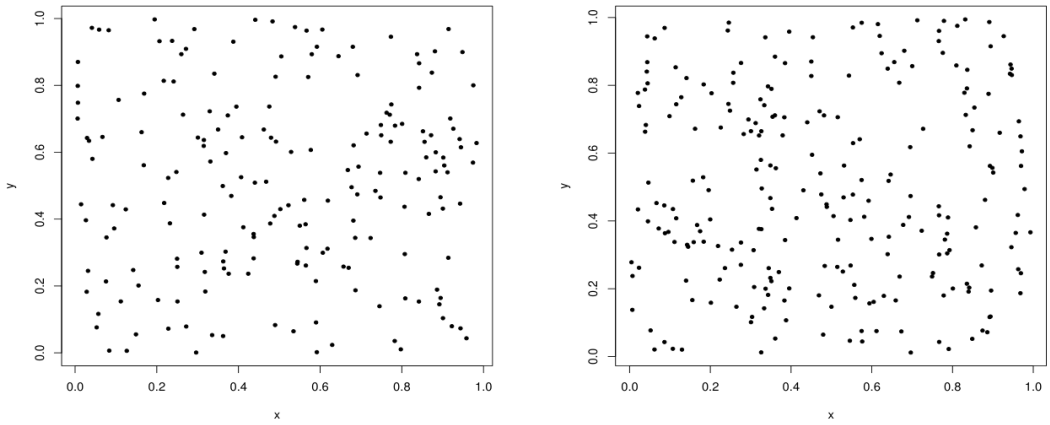


Figure 1: Spatial coordinates for the first two temporal instants.

```
corrmodel="Exp_Exp"
scale_s=0.3/3
scale_t=4/3
param=list(mean=mean,mean1=mean1,sill=sill,nugget=nugget,
           scale_s=scale_s,scale_t=scale_t)
```

We are now ready to simulate the space-time Gaussian RF using the function `GeoSim`:

```
ss1=GeoSim(coordx_dyn=coordx_dyn,coordt=coordt,
           corrmodel=corrmodel,X=X,model=model,param=param)$data
```

The simulation is performed using Cholesky decomposition. Note that the option `coordx_dyn` allows the specification of dynamical spatial coordinates as a list.

Estimation of Gaussian space-time random fields

Let us assume that we observe the RF Y at a finite set of spatial locations changing over time, i.e., $(\boldsymbol{s}_{i(l)}, t_l)$ with $l = 1, \ldots, T$ and $i(l) = 1, \ldots, N_l$. Let $f_Y(y_{il}, y_{jk})$ be the Gaussian density of the bivariate random vector $(Y(\boldsymbol{s}_{i(l)}, t_l), Y(\boldsymbol{s}_{j(k)}, t_k))^\top$. Then, the pairwise log-likelihood function is defined as

$$\mathrm{pl}(\boldsymbol{\theta}) = \sum_{(i,j,l,k)\in\mathcal{D}} \log\{f_Y(y_{il}, y_{jk})\}\, w_{ijlk}, \tag{3}$$

where $\mathcal{D}$ is a suitable set of indices for distinct observation pairs and w_{ijlk} are nonnegative weights not depending on $\boldsymbol{\theta}$.

The weights are specified as

$$w_{ijlk} = \begin{cases} 1, & \|\boldsymbol{s}_{i(l)} - \boldsymbol{s}_{j(k)}\| \leq d_s, \quad |t_l - t_k| \leq d_t, \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

where d_s and d_t are the spatial and temporal cutoffs, respectively. In this case $\boldsymbol{\theta} = (\beta_1, \beta_2, \sigma^2, \alpha_s, \alpha_t)^\top$. The pairwise likelihood estimator $\widehat{\boldsymbol{\theta}}_{\text{pl}}$ is obtained by maximizing (3) with respect to $\boldsymbol{\theta}$. In the `GeoModels` package we can choose the fixed parameters and the parameters that must be estimated. Pairwise likelihood estimation is performed with the function `GeoFit`:

```
# estimation with pairwise likelihood
fixed=list(nugget=nugget)
start=list(mean=mean,mean1=mean1,sill=sill,
           scale_s=scale_s,scale_t=scale_t)
fit=GeoFit(data=ss1,coordx_dyn=coordx_dyn,
           coordt=coordt,corrmodel=corrmodel,
           maxdist=0.1,maxtime=1,X=X,optimizer="BFGS",
           start=start,fixed=fixed,model=model)
```

The object `fit` includes information about the pairwise likelihood estimation:

```
fit
 Maximum Composite-Likelihood Fitting of Gaussian Random Fields
 Setting : Marginal Composite-Likelihood
 Model : Gaussian
 Type of the likelihood objects : Pairwise
 Covariance model : Exp_Exp
 Optimizer : BFGS
 Number of spatial coordinates : 3273
 Number of dependent temporal realisations : 16
 Type of the random field : univariate
 Number of estimated parameters : 5
 Type of convergence : Successful
 Maximum log-Composite-Likelihood value : -75860.09
 Estimated parameters :
    mean    mean1  scale_s  scale_t    sill
  0.3827 -0.2541   0.0889   1.2870  0.9268
```

Note that `maxdist=0.1` and `maxtime=1` specify the cutoffs of the weight function (4), i.e., $d_s = 0.1$ and $d_t = 1$; they do not specify the range parameters of the correlation function.

Checking model assumptions

Given the estimates of the regression and sill parameters $\widehat{\boldsymbol{\beta}}$, $\widehat{\sigma}^2$, the estimated residuals

$$\widehat{Z}(\boldsymbol{s}_{i(l)}, t_l) = \frac{Y(\boldsymbol{s}_{i(l)}, t_l) - X(\boldsymbol{s}_{i(l)}, t_l)^\top \widehat{\boldsymbol{\beta}}}{\sqrt{\widehat{\sigma}^2}}, \quad l = 1, \ldots, T, \ i = 1, \ldots, N_l,$$

can be viewed as a realization of a zero-mean, unit-variance stationary Gaussian RF with correlation function $\rho(\boldsymbol{h}, u)$. The residuals can be computed using the `GeoResiduals` function:

```
res=GeoResiduals(fit)  # computing residuals
```

Then the marginal distribution assumption on the residuals can be graphically checked with a Gaussian QQ-plot (Figure 2, left panel) using the function `GeoQQ`:

```
# checking model assumptions: marginal distribution
GeoQQ(res)
```

The correlation model assumption can be checked by comparing the empirical and estimated space-time semivariogram functions using `GeoVariogram` and `GeoCovariogram` (Figure 2, right panel):

```
# checking model assumptions: ST variogram model
vario=GeoVariogram(data=res$data,coordx_dyn=coordx_dyn,
         coordt=coordt,maxdist=0.6,maxtime=7)
GeoCovariogram(res,vario=vario,fix.lagt=1,fix.lags=1,
         show.vario=TRUE,pch=20)
```

We remark that the empirical space-time semivariogram computation is slightly modified with respect to the classical fixed-location version because no exact repeated spatial locations are available for directly estimating the purely temporal semivariogram at zero spatial lag.

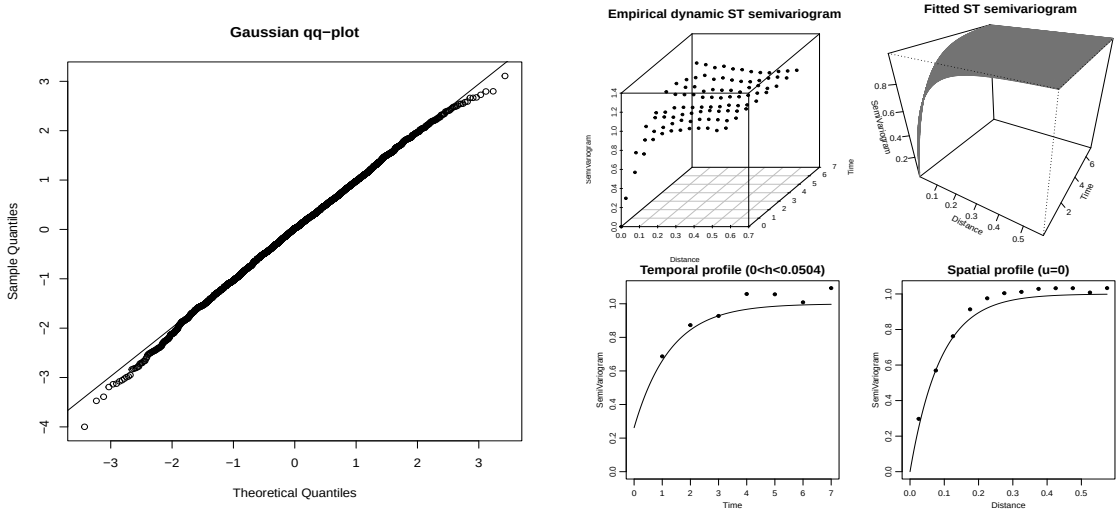


Figure 2: Left: QQ-plot for the residuals of the space-time Gaussian RF. Right: space-time empirical vs. estimated semivariogram function for the residuals.

Prediction of space-time Gaussian random fields

For a given space-time location $(\boldsymbol{s}_0, t_0)$ with associated covariates $X(\boldsymbol{s}_0, t_0)$, the optimal Gaussian linear prediction is computed as:

$$\widehat{Y}(\boldsymbol{s}_0, t_0) = X(\boldsymbol{s}_0, t_0)^\top \widehat{\boldsymbol{\beta}} + \sum_{l=1}^{T} \sum_{i=1}^{N_l} \lambda_{l,i} \left[Y(\boldsymbol{s}_{i(l)}, t_l) - X(\boldsymbol{s}_{i(l)}, t_l)^\top \widehat{\boldsymbol{\beta}} \right]. \tag{5}$$

The weight vector $\boldsymbol{\lambda} = (\lambda_{1,1}, \ldots, \lambda_{T,N_T})^\top$ is given by $\boldsymbol{\lambda} = R^{-1}\boldsymbol{c}$, where $\boldsymbol{c}$ contains the correlations between $Y(\boldsymbol{s}_0, t_0)$ and all observations, ordered by time and then site. The $N \times N$ correlation matrix ($N = \sum_l N_l$) has entries

$$R_{(l,i),(k,j)} = \text{cor}\{Y(\boldsymbol{s}_{i(l)}, t_l), Y(\boldsymbol{s}_{j(k)}, t_k)\}, \quad i = 1, \ldots, N_l, \ j = 1, \ldots, N_k.$$

Kriging can be performed using the `GeoKrig` function. We just need to specify the spatial locations and temporal instants to predict. In this example we consider a spatial regular grid and the first two temporal instants:

```
# spatial locations to predict
xx=seq(0,1,0.02)
loc_to_pred=as.matrix(expand.grid(xx,xx))
times=c(coordt[1],coordt[2])
```

Additionally, we need to specify the associated covariates:

```
Nloc=nrow(loc_to_pred)*length(times)
# Xloc is ordered by location, with both times for each site.
Xloc=cbind(rep(1,Nloc),runif(Nloc))
```

Then the optimal linear prediction (5), using the estimated parameters, can be performed with `GeoKrig`:

```
param_est=as.list(c(fit$param,fixed))
pr=GeoKrig(data=ss1,coordx_dyn=coordx_dyn,coordt=coordt,
           corrmodel=corrmodel,X=X,Xloc=Xloc,model=model,
           loc=loc_to_pred,time=times,param=param_est)
```

A kriging map for the first two temporal instants (Figure 3), with a comparison with observed data, can be obtained with the following code:

```
par(mfrow=c(2,2))
i=1
for(i in 1:length(times)) {
  quilt.plot(coordx_dyn[[i]],ss1[[i]])
  image.plot(xx,xx,matrix(pr$pred[i,],ncol=length(xx)),xlab="",
        main=paste(" Kriging Time=",times[i]),ylab="")
}
```

References

Moreno Bevilacqua, Christian Caamaño Carrillo, Francisco Cuevas-Pacheco, and Víctor Morales-Oñate. *GeoModels: Procedures for Gaussian and Non Gaussian Geostatistical (Large) Data Analysis*, 2026. URL `https://CRAN.R-project.org/package=GeoModels`. R package version 2.3.0.

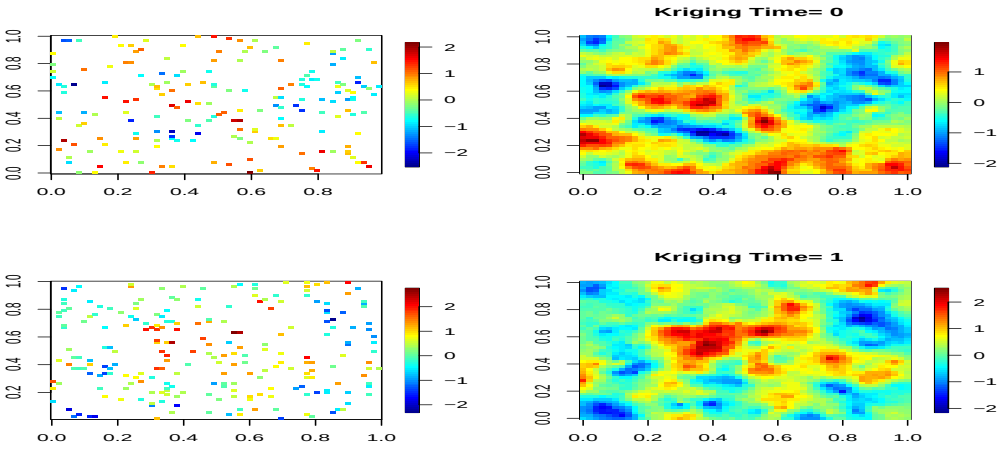


Figure 3: Gaussian space-time kriging for the first two temporal instants.